MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, MODULAR, CONNECTOR ASSEMBLY, TYPE V, 40-PIN, TWO PIECE INSERT, AND 20-PIN, ONE HALF, ONE PIECE INSERT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connector assemblies described herein shall consist of this specification and the latest issue of MIL-C-28754.

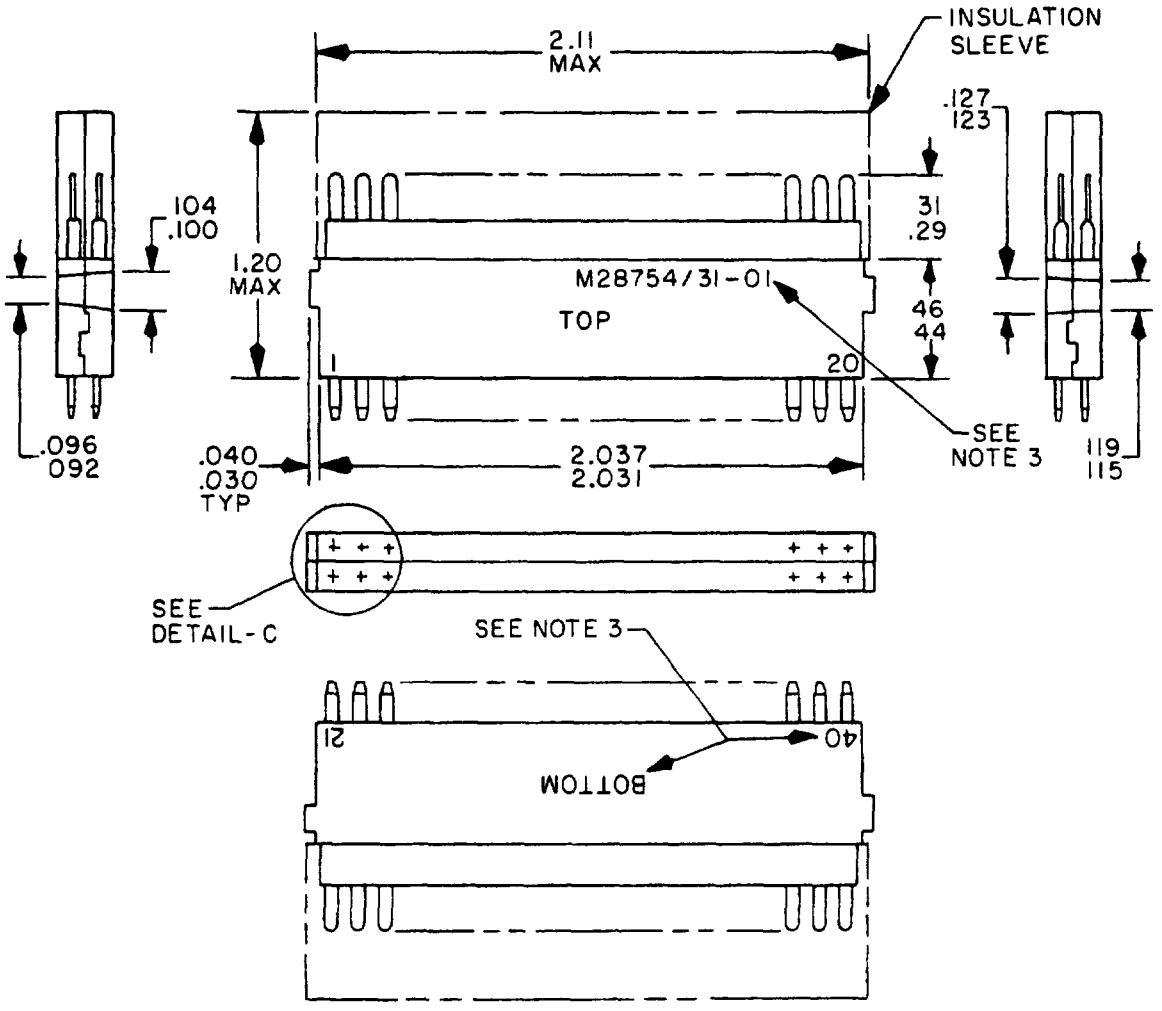


FIGURE 1A. Top and bottom inserts.

FIGURE 1. Dimensions and configurations

© denotes changes

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 202610-20T14:39Z
Check the source to verify that this is the current version before use.

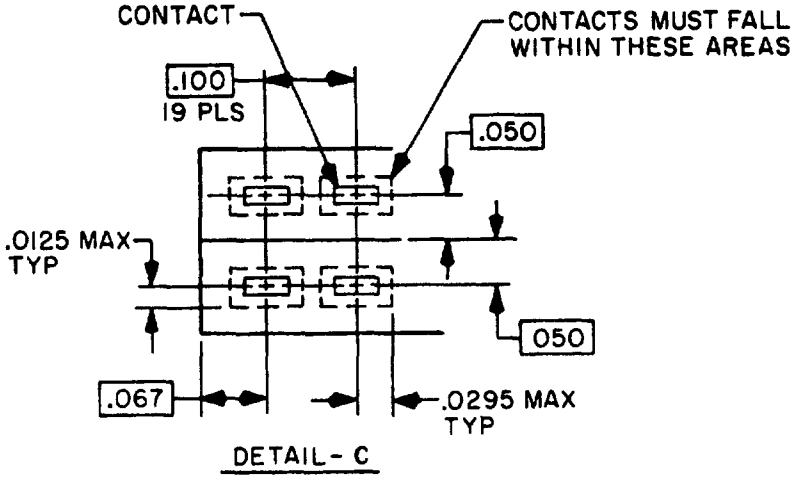


FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T14:39Z
Check the source to verify that this is the current version before use.

INCHES	MM
.005	.13
.01	.3
.0125	.318
.0295	.749
.030	.76
.040	1.02
.050	1.27
.067	1.70
.092	2.34
.096	2.44
.100	2.54
.104	2.64
.115	2.92
.119	3.02
.123	3.12
.127	3.23
.29	7.4
.30	7.6
.31	7.9
.44	11.2
.46	11.7
1.20	30.5
2.031	51.59
2.037	51.74
2.11	53.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Mark in this area in white characters, .30 inch high maximum.
4. Individual inserts shall be marked in this area. When combined to make a two-piece assembly, the marking shall be per figure 1.
5. Unless otherwise specified, tolerance is ±.005 for three place decimals, ±.01 for two place decimals and ±2° on angles.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:39Z
Check the source to verify that this is the current version before use.

© TABLE I. Dash numbers and applicable insert.

Dash no.	Top insert MIL-C-28754/50	Bottom insert MIL-C-28754/51	Figure
01	-01	-01	1A
02	-01	-02, -04	1A
03	-01	-03	1A
04	-02, -04	-01	1A
05	-02, -04	-02, -04	1A
06	-02, -04	-03	1A
07	-03	-01	1A
08	-03	-02, -04	1A
09	-03	-03	1A

REQUIREMENTS:

Dimensions and configuration: See figure 1 and applicable MIL-C-28754 specification sheet (see table I).

Design characteristics: See table II.

Contacts: In accordance with applicable MIL-C-28754 specification sheet (see table I).

Material: In accordance with applicable MIL-C-28754 specification sheet (see table I).

Temperature rating: -55°C to +105°C.

Voltage rating: 300 volts, AC (rms) at sea level.

Solderability: The insulation sleeve is to be removed prior to testing. For qualification testing, solderability to be transferred from subgroup 1 to subgroup IV of table V, MIL-C-28754. Number of units to be inspected shall be 2; number of defects allowed shall be 0.

Marking: In accordance with MIL-C-28754 and figure 1.

Military part number M28754/31- (dash number from table I).

TABLE II. Design characteristics.

Conductor size (AWG)	Current rating (Amps)	Contact resistance (MV)	Durability (MV)	Solder joint resistance (MV)
20	3	20	20	4.0
22	3	20	20	4.0
24	3	20	20	4.0
26	2	13.3	13.3	4.0
28	1.5	10.0	10.0	4.0
30	1.0	6.7	6.7	4.0

Source: https://assist.dla.mil -- Downloaded: 202610-20T14:39Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities
Navy - MC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, MI
Navy - YD

Agent.
DLA - ES

Preparing activity:
Navy - EC

(Project 5935-3266-27)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-C-28754 / 31C

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO

DATE

DD FORM 1 OCT 78 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 202610-20T14:39Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, D. C. 20363

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20363 .

ATTN: ELEX 8111

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:39Z
Check the source to verify that this is the current version before use.